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6. You’re now ready to code with us!



PyIU Python Crash Course

AN INTRODUCTION TO PYTHON FOR STUDENTS INTERESTED IN ASTRONOMY/STEM

By: IU Astronomy & Physics Graduate Students

09/16/26

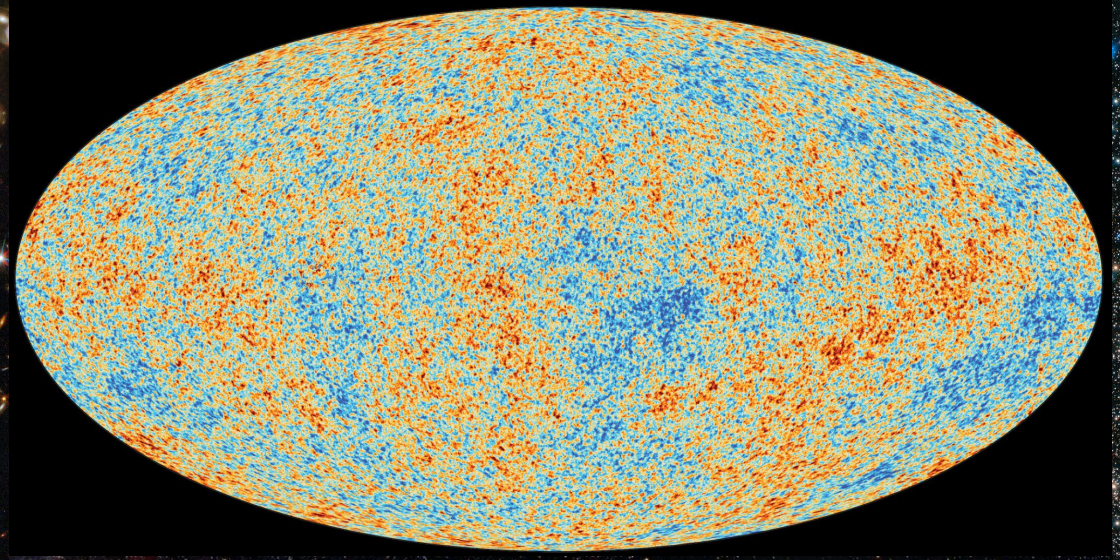
Crash Course Purpose

- Astronomers, physicists, and other scientists use **programming** as a tool to analyze data for research
- Python has become the most-used programming language, becoming a vital skill!



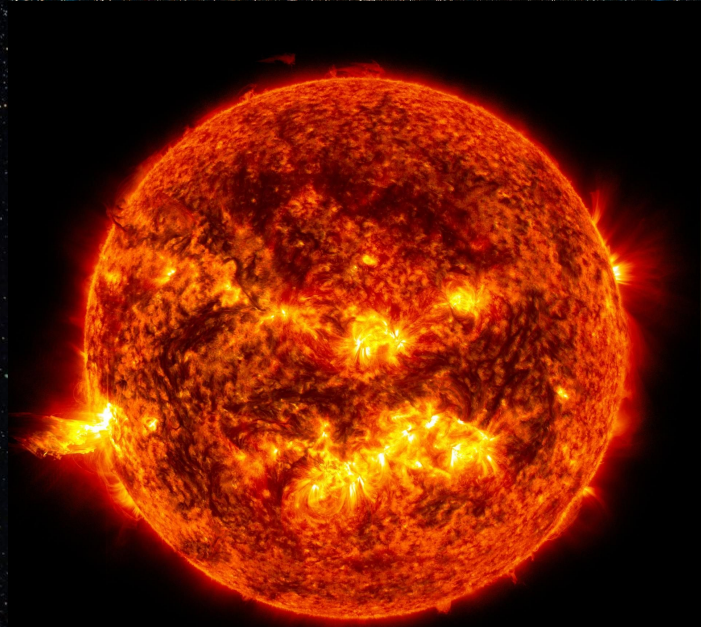
But Python is much more user-friendly than these!
(at the cost of being a bit slower)

Other common languages:
C, C++, C#
Java & JavaScript
SQL
R
MATLAB
Rust
Perl
HTML & CSS
Fortran
Julia

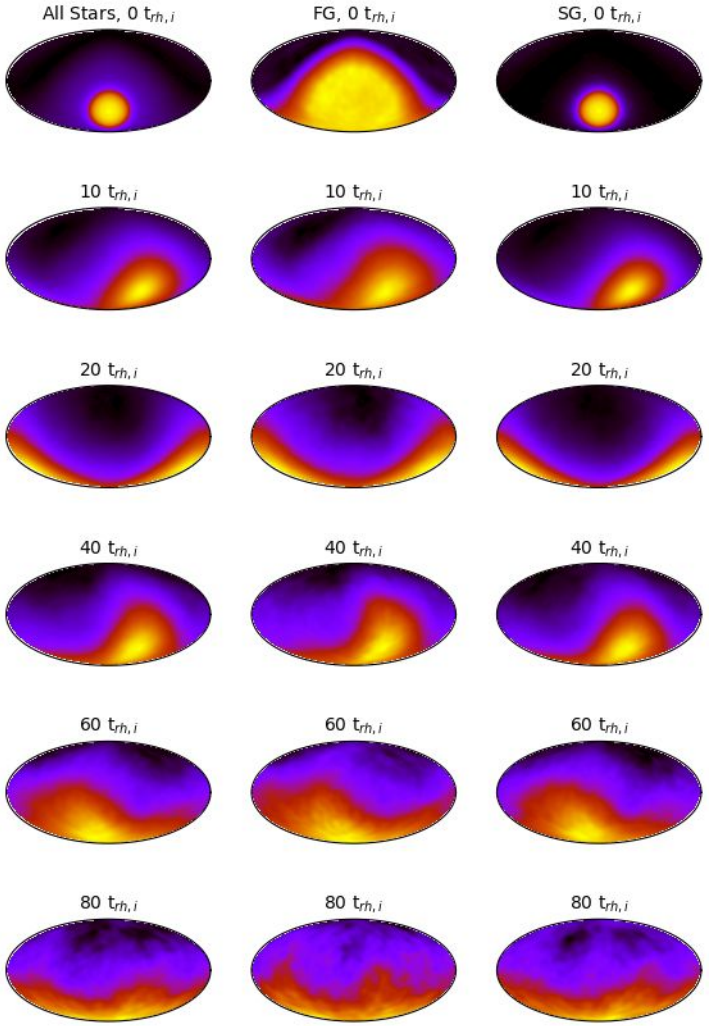



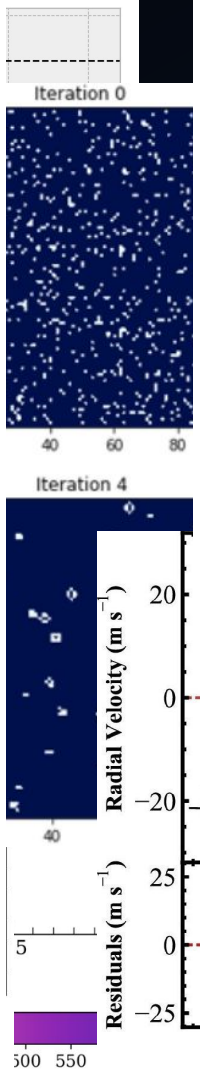

Coding is your gateway to the universe...

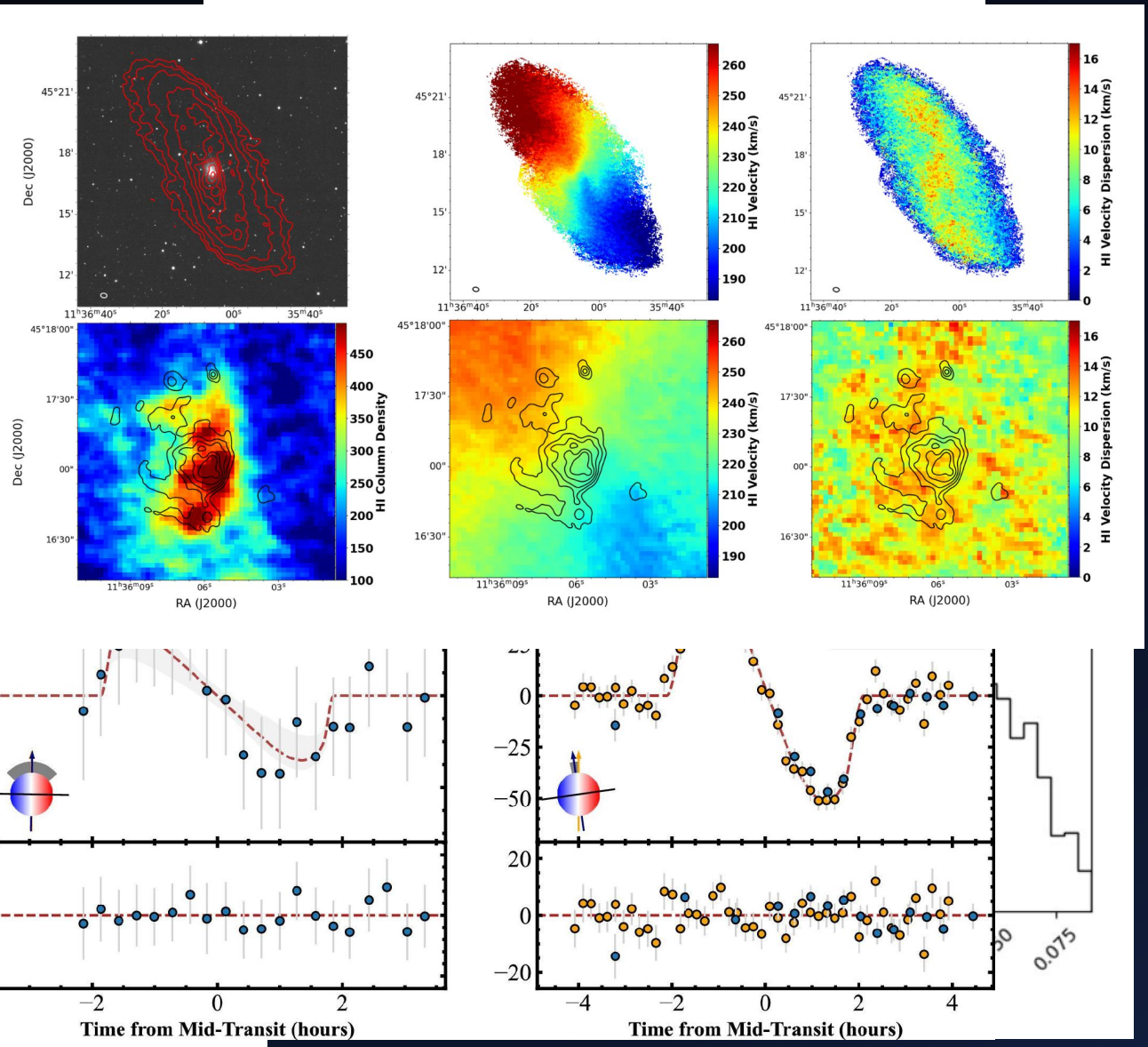




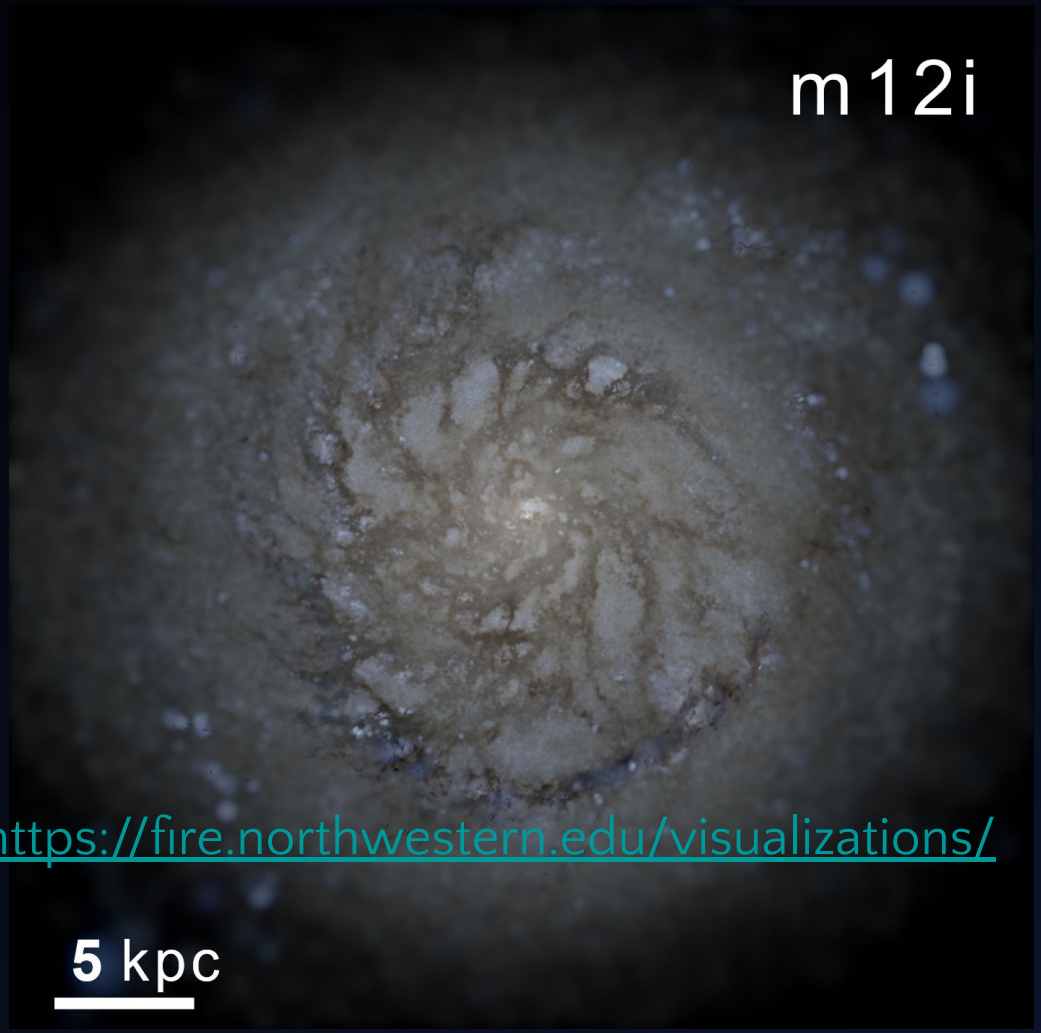
What we've done with Python!



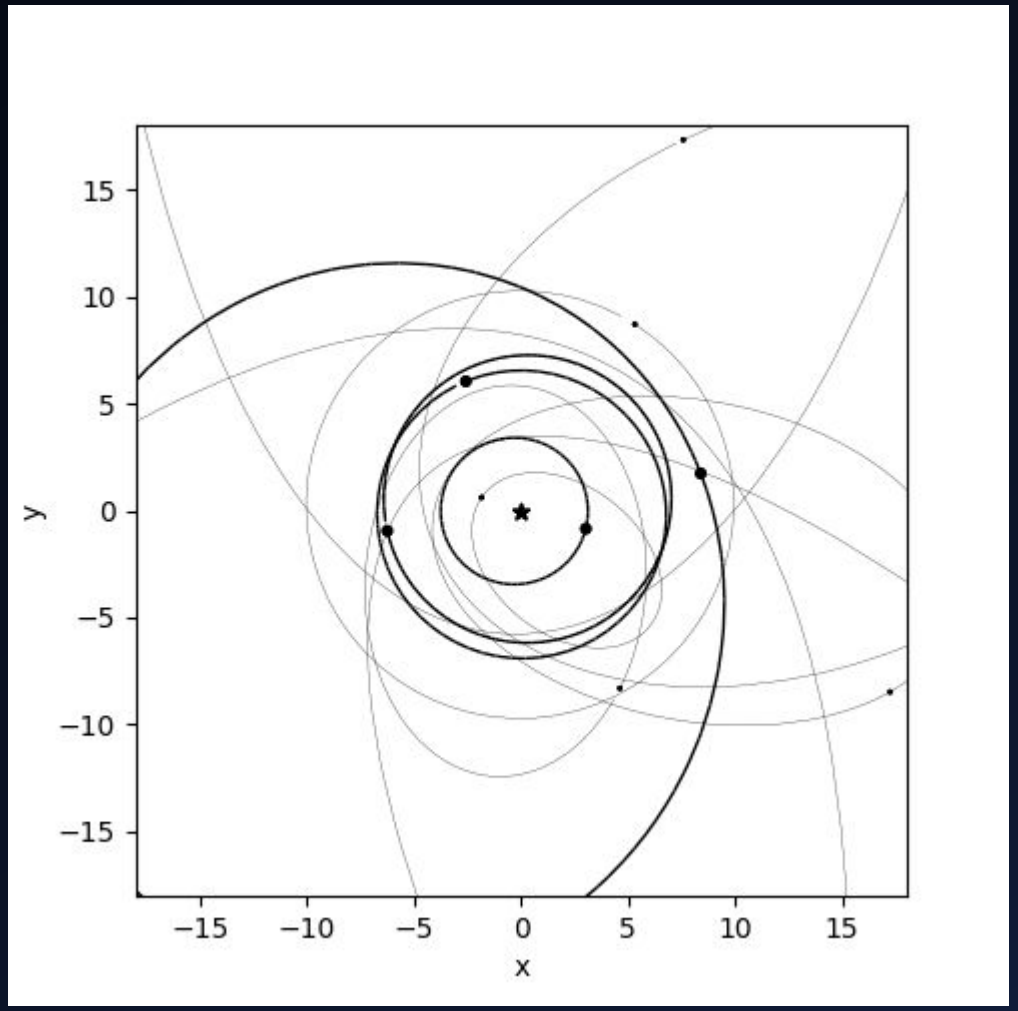




More Examples of Python from IU Astronomy



https://fire.northwestern.edu/visualizations/



Python



Pros of Python:

- Simple and **user-friendly**
- Interpreted, high-level language

Cons:

- A bit of a black box, **slow**
- Interpreted, high-level language

Documentation:
https://docs.python.org/3/

Lots of “layers” to executing the code, computes line-by-line

Easily readable to humans… not so much for a computer

A Demonstrative Example: “Hello World”





Where to write code in Python



Where to write code in Python



Where to write code in Python

- IDEs: Integrated Development Environments





PyCharm

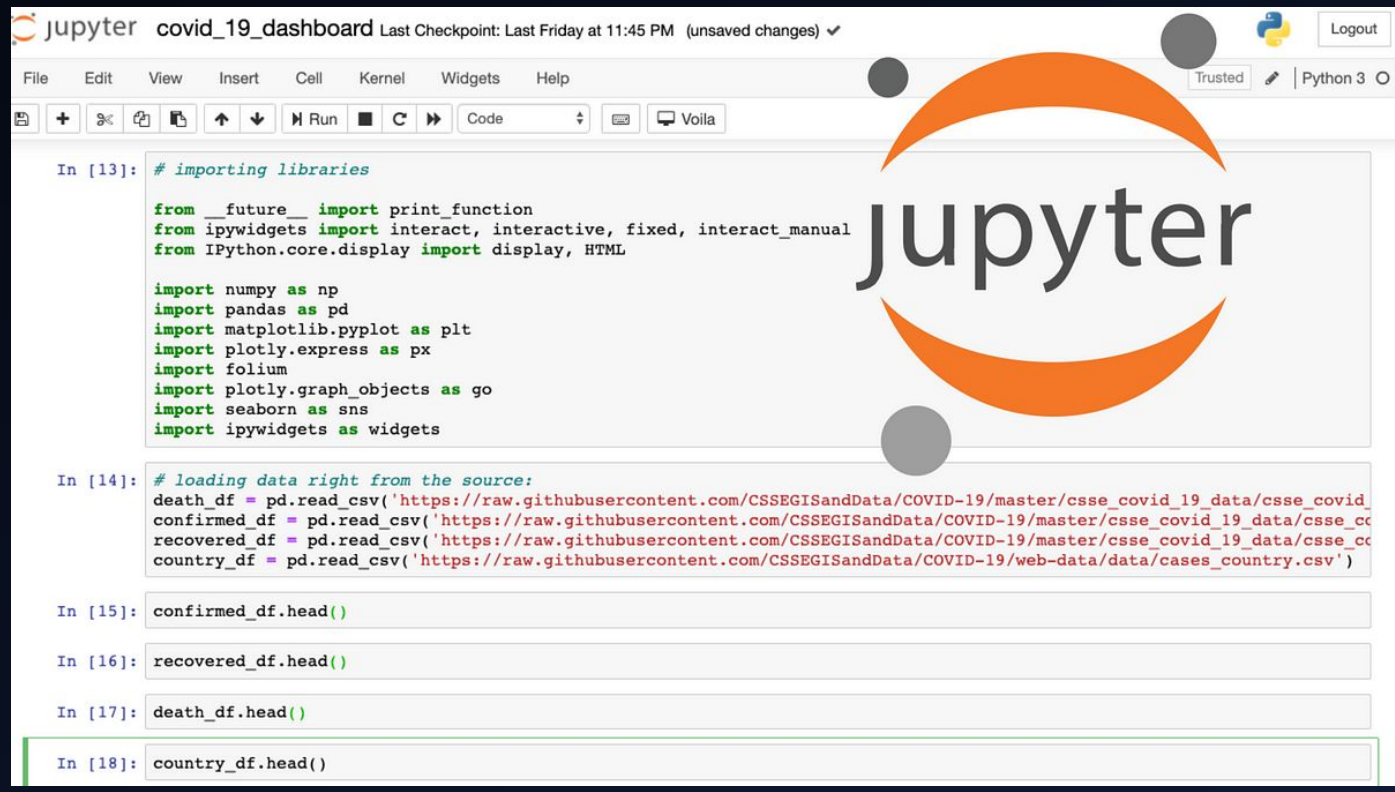




VSCode

What we're using for the Crash Course

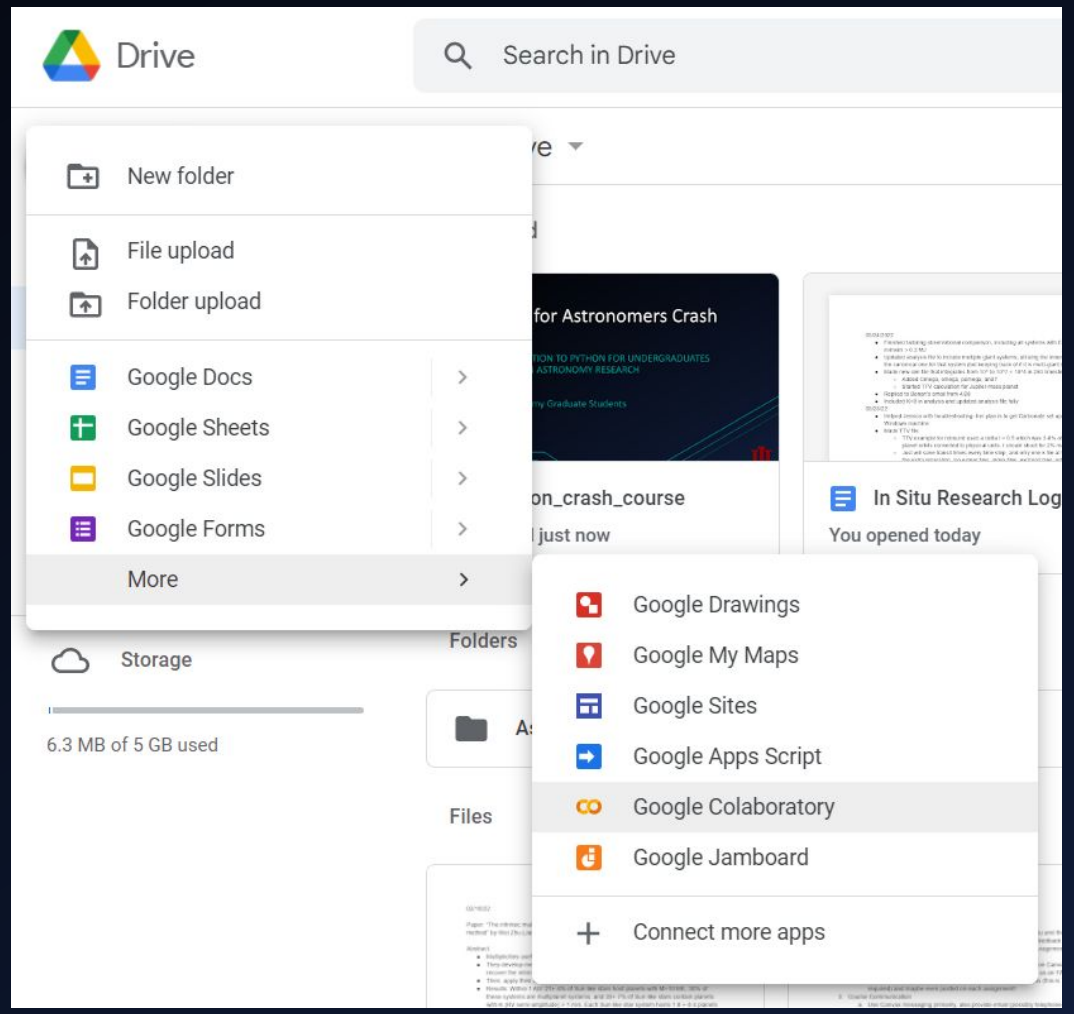
- **Jupyter notebook**: web-based interactive Python development tool
- **Cell-by-cell code blocks** allow for:
  - Easy debugging & visualization
  - Quick and efficient code development



How we're accessing Jupyter: Google "Colab"oratory

colab.research.google.com

- Free web-based access to Jupyter notebooks
- Contains searchable practice examples and datasets
- Allows collaboration* **(Google docs but for coding)**
  - *asynchronously



The Artemis II Mission

PyIU Python Crash Course





The Goal

- Establish a permanent human presence on the Moon
- Prepare for a crewed mission to Mars
- Artemis II was a test of NASA's capability to send a crewed vessel to the Moon and back



Image Credit: NASA

The Astronauts

- Christina Koch (NASA): Mission Specialist 1
- Victor Glover (NASA): Pilot
- Jeremy Hansen (CSA): Mission Specialist 2
- Reid Wiseman (NASA): Commander



Image Credit: NASA

The Launch

- Launched from Kennedy Space Center in Florida on April 1, 2026, at 6:35 pm EDT
- Launch video!

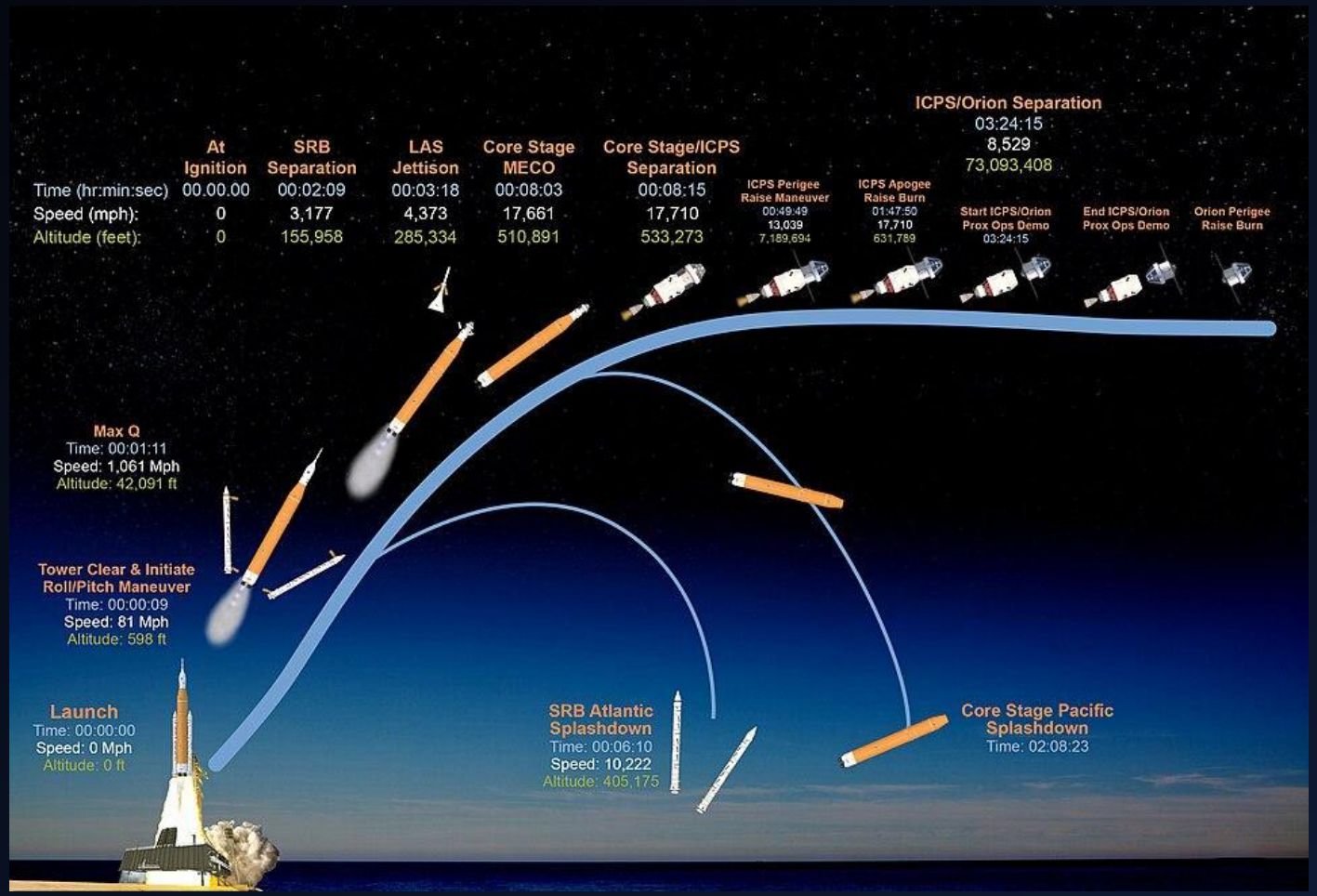


Image Credit: NASA

The Journey

- Orion capsule flew around the Moon and back in 9 days, traveling around 700,000 miles (~1,100,000 km)
- Astronauts took lots of beautiful pictures throughout their journey

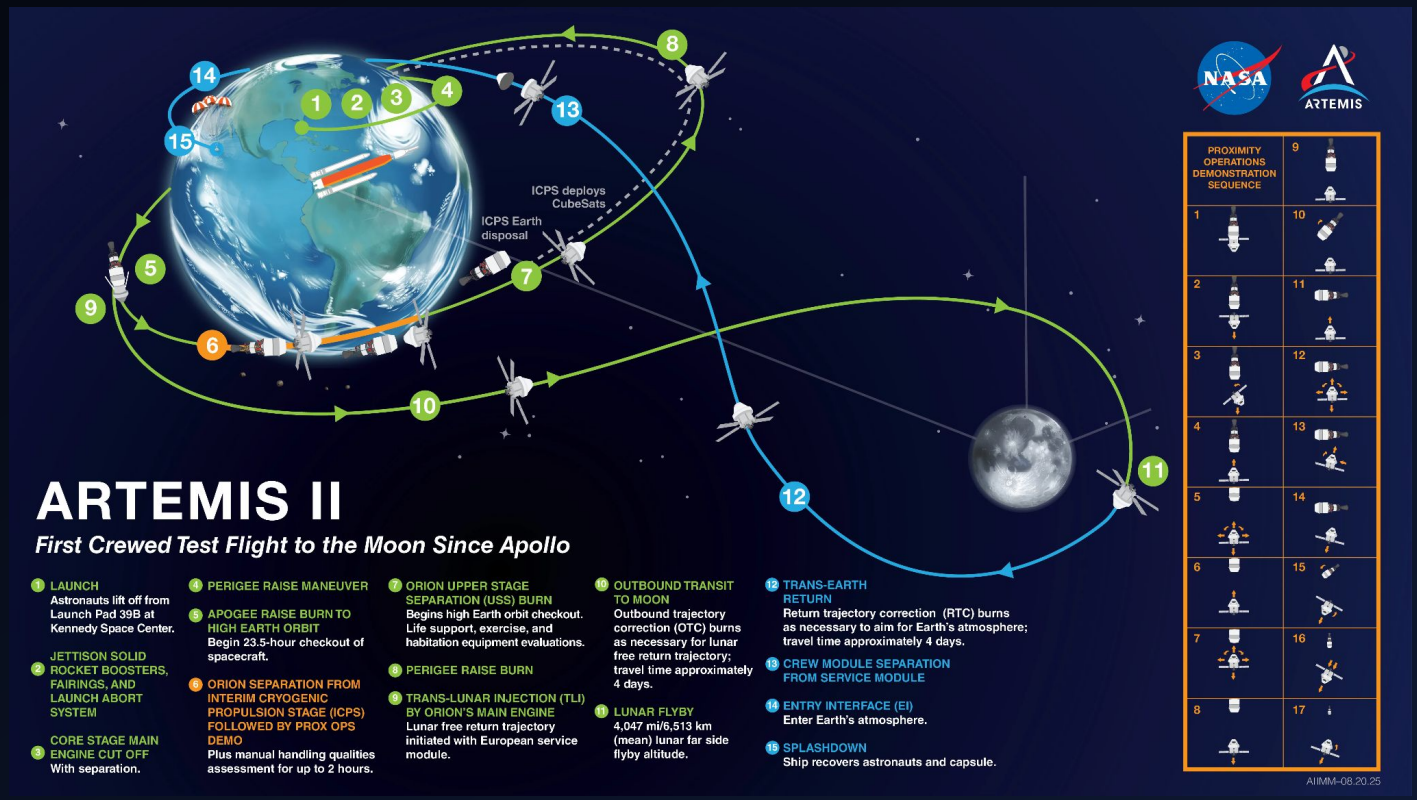


Image Credit: NASA

The Journey



Image Credit: NASA

The Journey



Image Credit: NASA

Your Turn

- Incrementally apply Python fundamentals
- Create your own visualization of the Artemis II mission's trajectory using real NASA data!



Image Credit: NASA

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Let’s Code!



Advanced Python Workshop: Coming this November!

- Learn techniques that you can apply directly to your **research/upper-level coursework**
- Casual, highly interactive environment with experienced instructors
- Potential topics:
  - Python installation/environments
  - Machine learning
  - Advanced plotting
  - Statistics
  - Numerical integration & simulations
  - Big data analysis & data fitting
  - **YOU tell US!**

